

Substituição

$$H(x) = \int f(g(x))g'(x) \, dx + k$$

de modo que

$$\int_a^b f(g(x))g'(x) \, dx = H(b) - H(a)$$

Também

$$\int_a^b f(g(x))g'(x) \, dx = \int_{g(a)}^{g(b)} f(t) \, dt$$

Partes

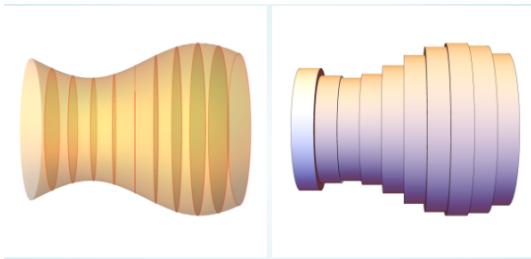
$$\int f(x)g(x) \, dx = f(x) \cdot G(x) - \int f'(x)G(x) \, dx$$

vale

$$\int_a^b f(x)g(x) \, dx = f(b) \cdot G(b) - f(a)G(a) - \int_a^b f'(x)G(x) \, dx$$

Volume

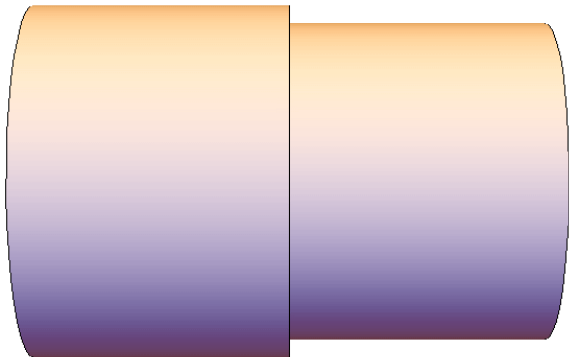
Volume por seção transversal



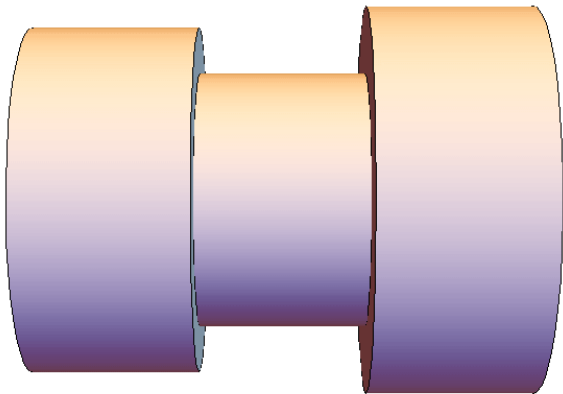
Volume por seção transversal



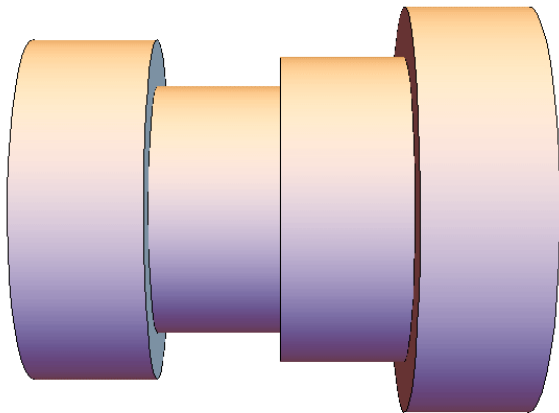
Volume por seção transversal



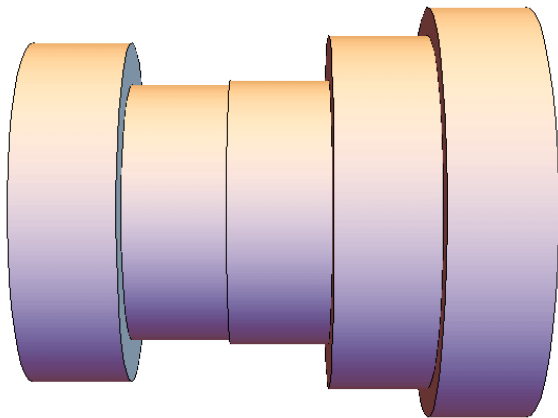
Volume por seção transversal



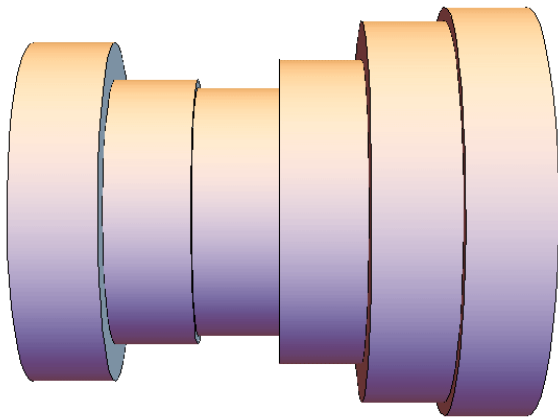
Volume por seção transversal



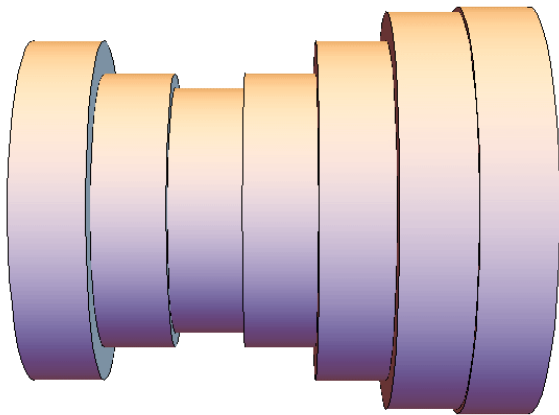
Volume por seção transversal



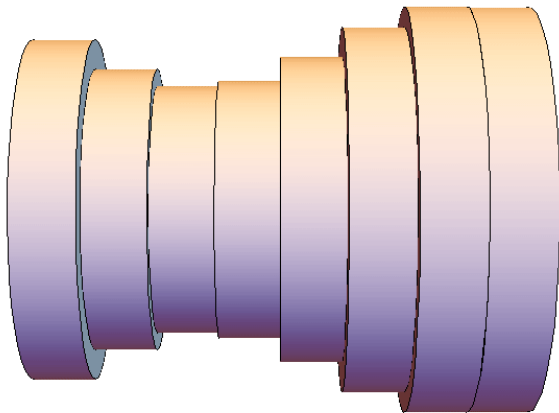
Volume por seção transversal



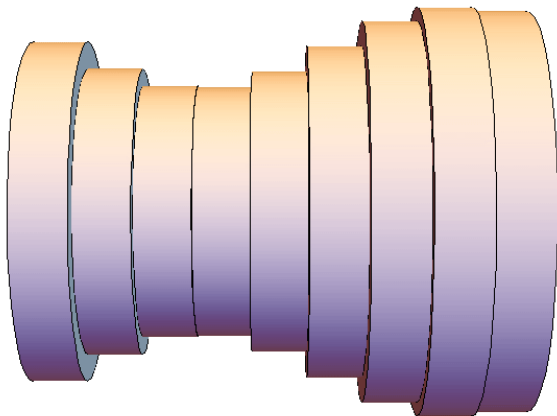
Volume por seção transversal



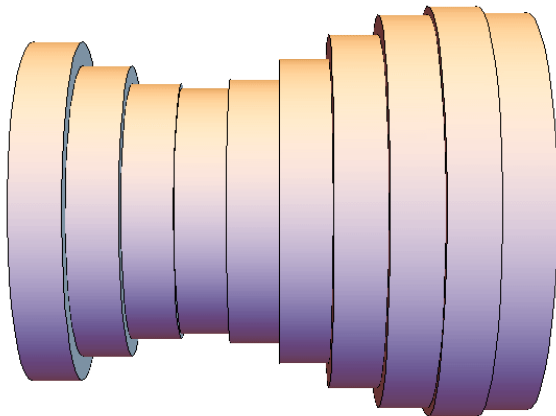
Volume por seção transversal



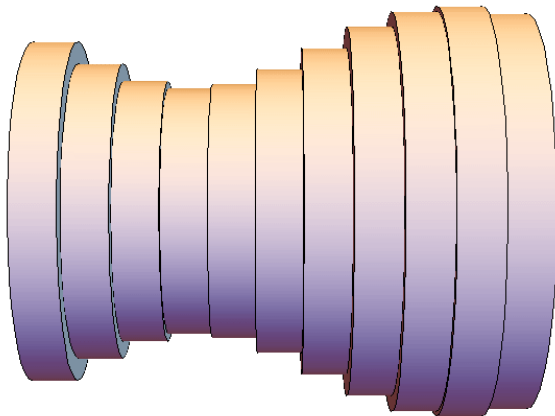
Volume por seção transversal



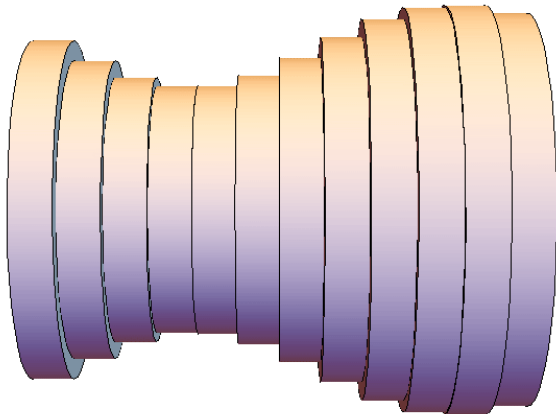
Volume por seção transversal



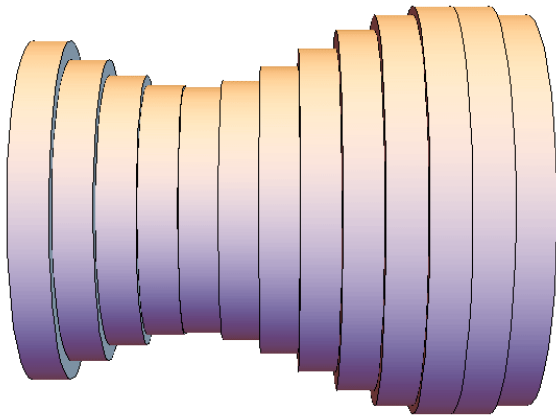
Volume por seção transversal



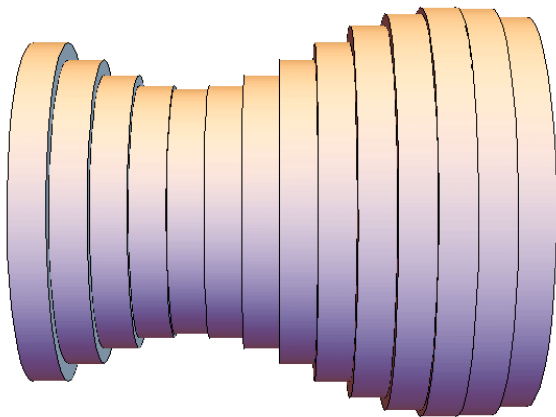
Volume por seção transversal



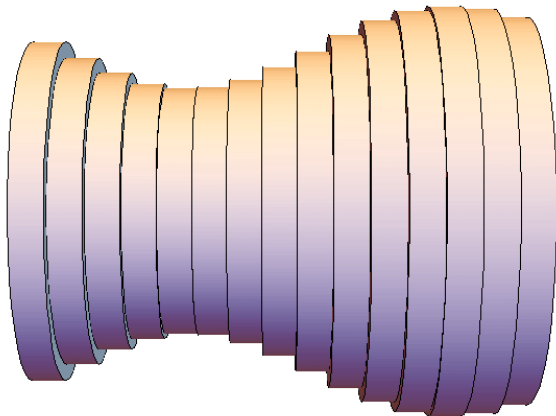
Volume por seção transversal



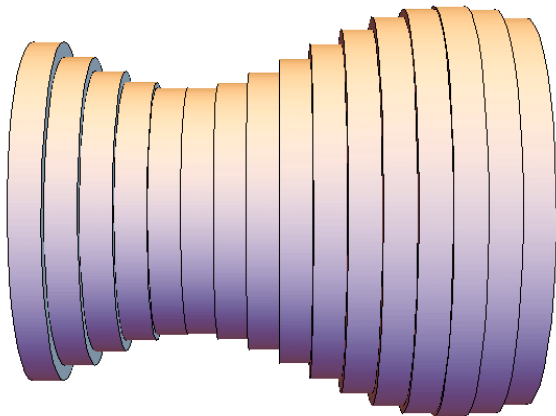
Volume por seção transversal



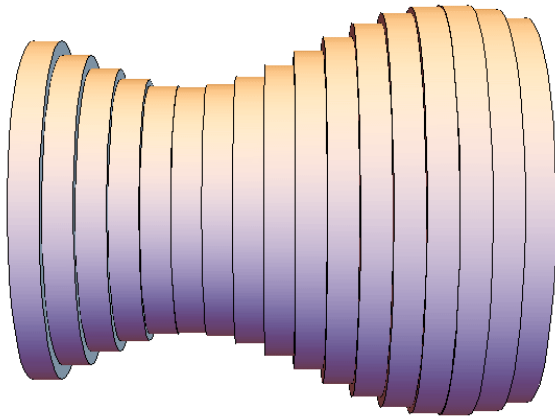
Volume por seção transversal



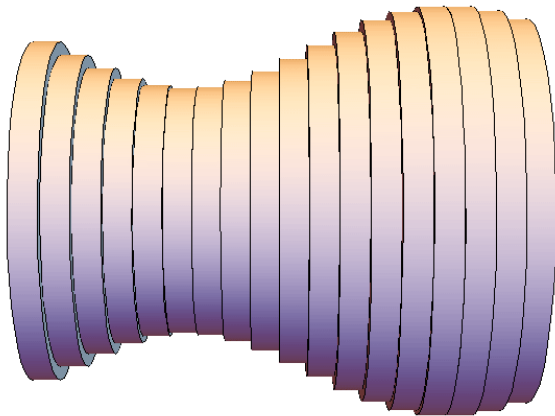
Volume por seção transversal



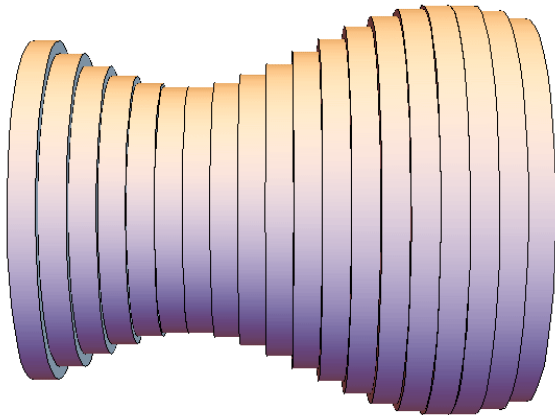
Volume por seção transversal



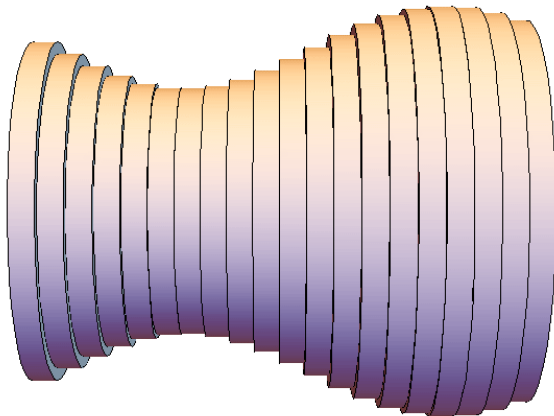
Volume por seção transversal



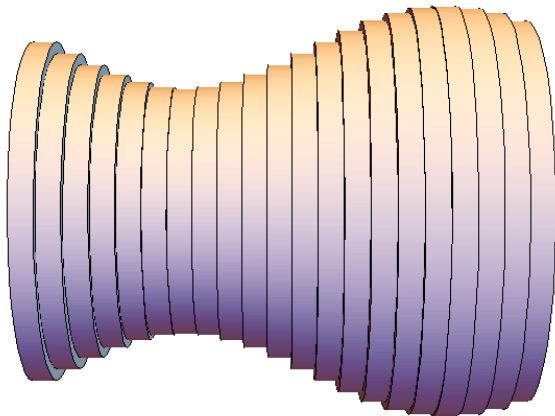
Volume por seção transversal



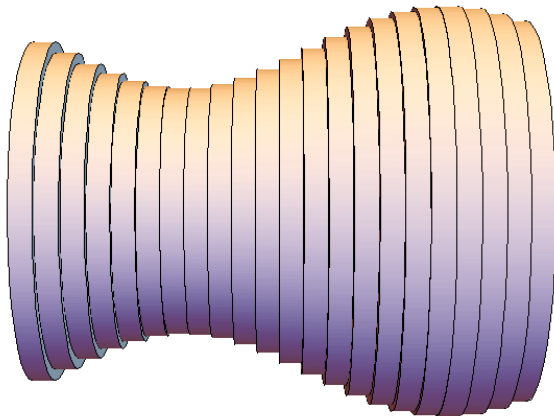
Volume por seção transversal



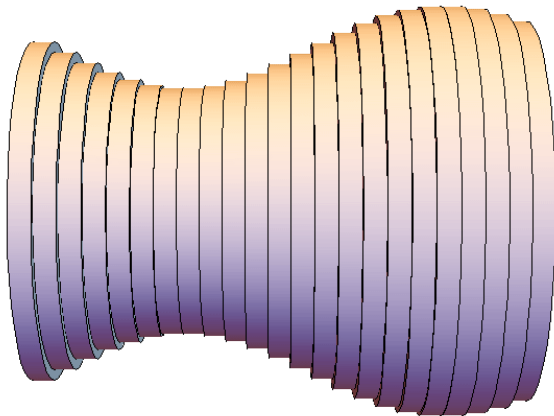
Volume por seção transversal



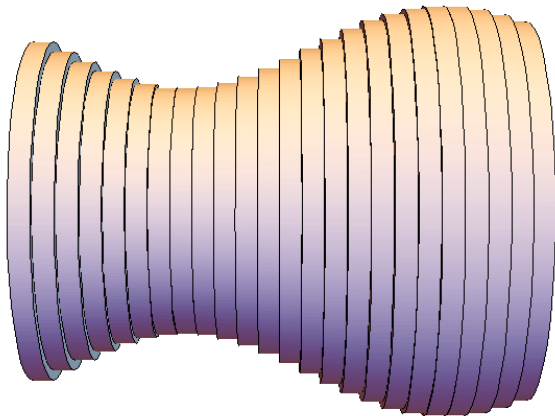
Volume por seção transversal



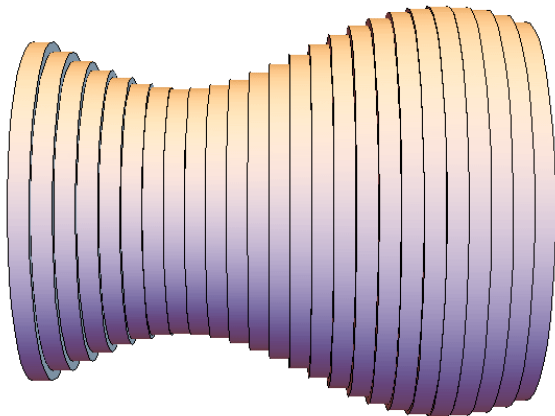
Volume por seção transversal



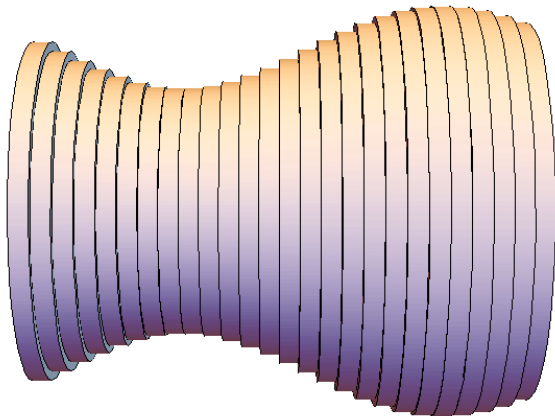
Volume por seção transversal



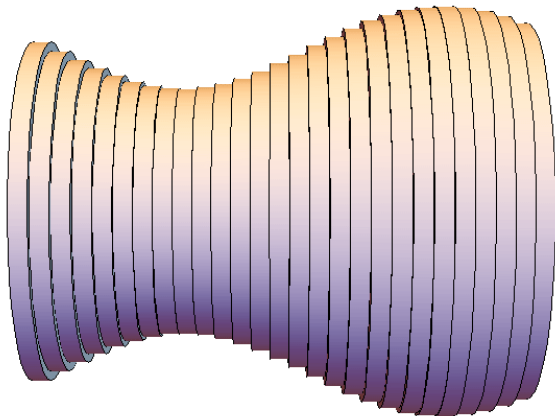
Volume por seção transversal



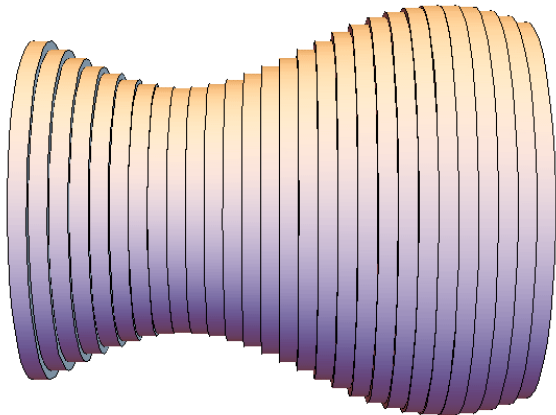
Volume por seção transversal



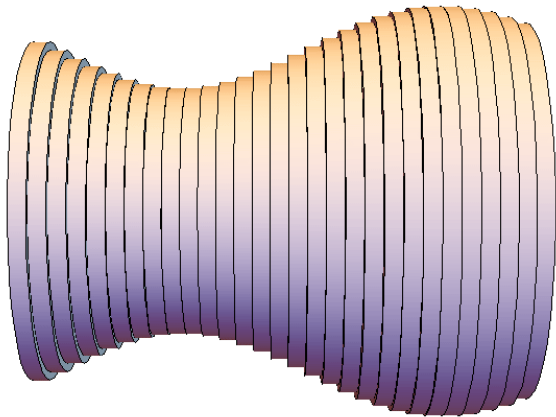
Volume por seção transversal



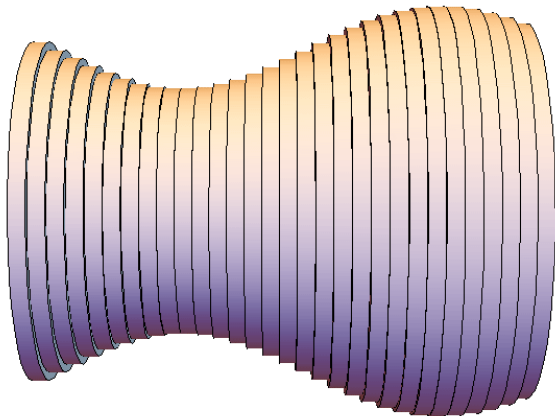
Volume por seção transversal



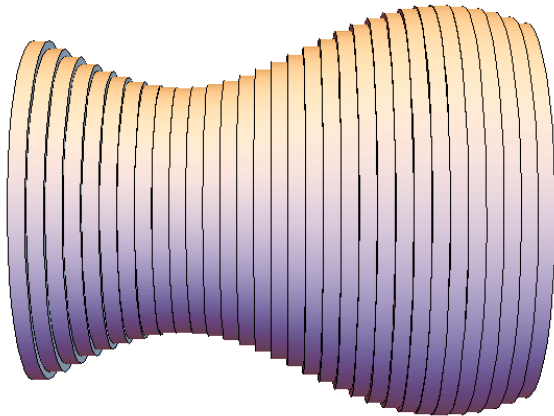
Volume por seção transversal



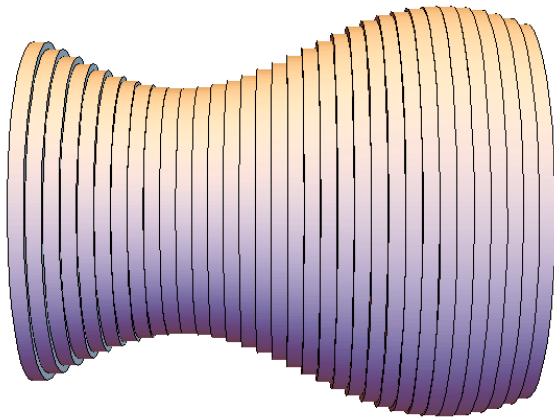
Volume por seção transversal



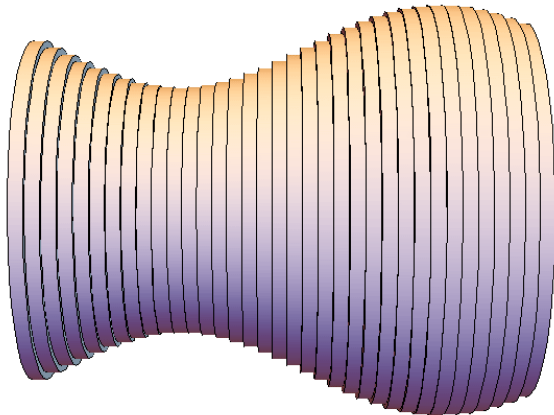
Volume por seção transversal



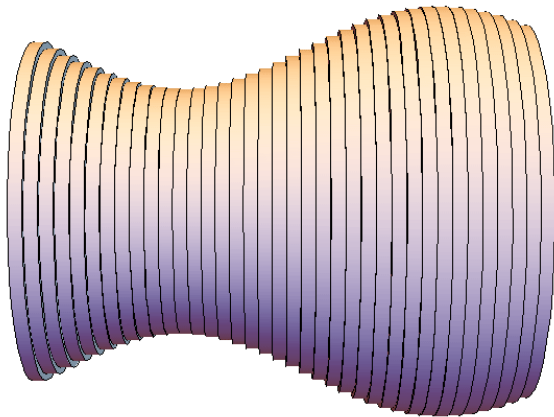
Volume por seção transversal



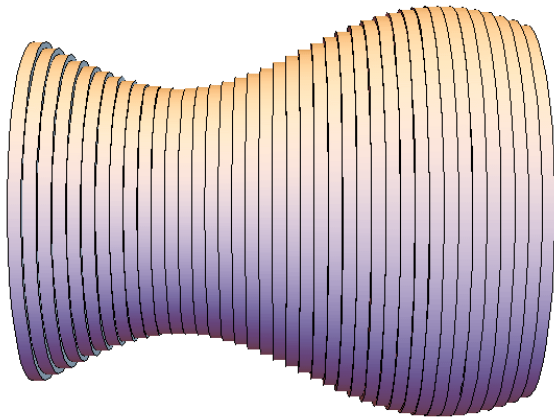
Volume por seção transversal



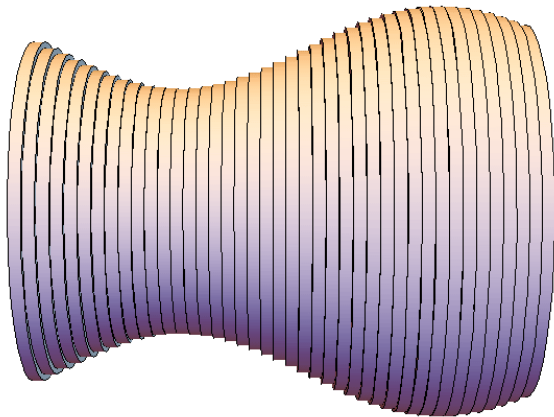
Volume por seção transversal



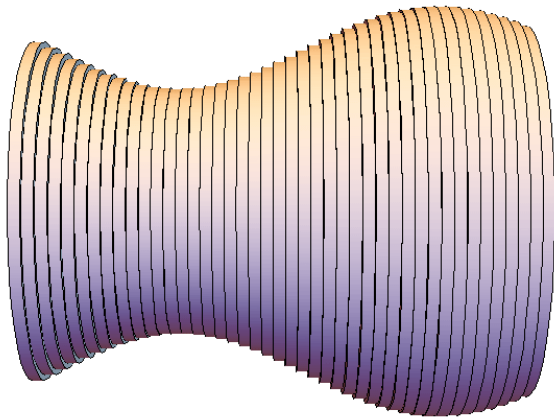
Volume por seção transversal



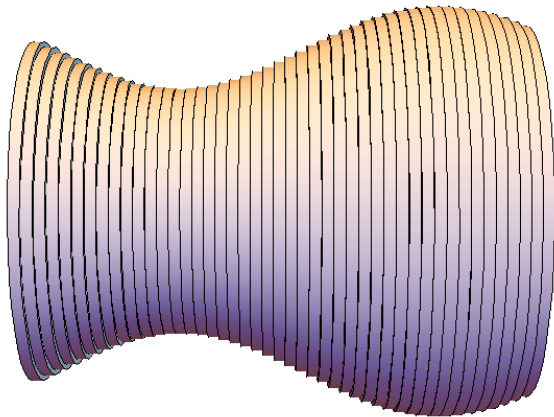
Volume por seção transversal



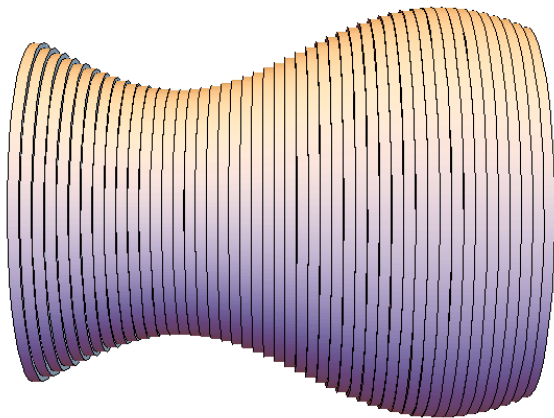
Volume por seção transversal



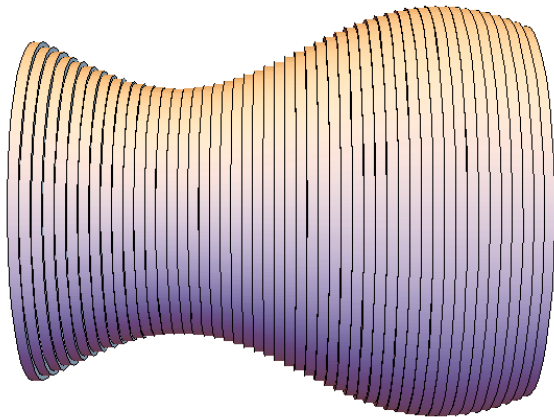
Volume por seção transversal



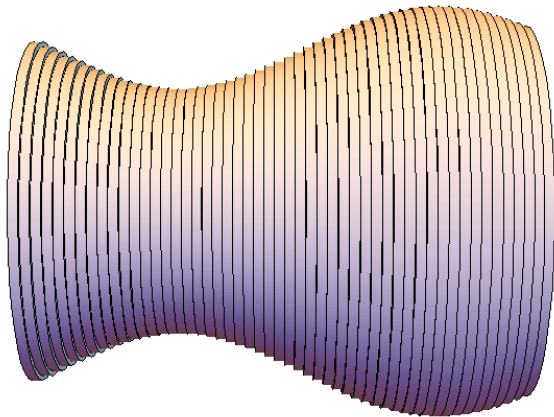
Volume por seção transversal



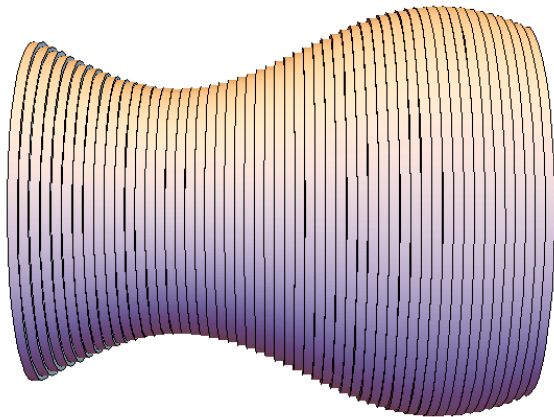
Volume por seção transversal



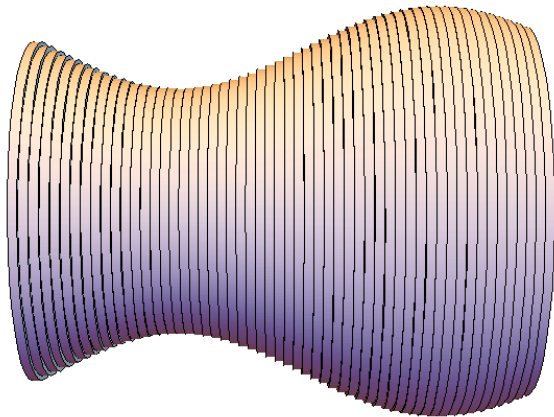
Volume por seção transversal



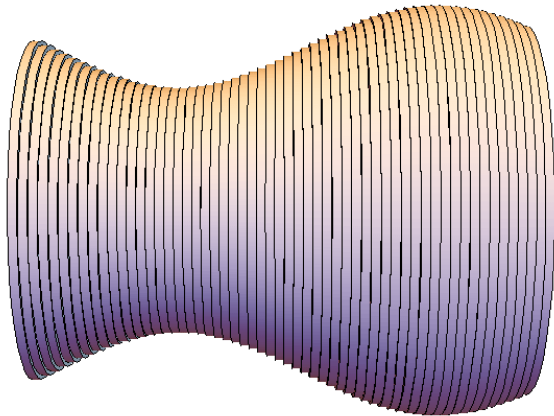
Volume por seção transversal



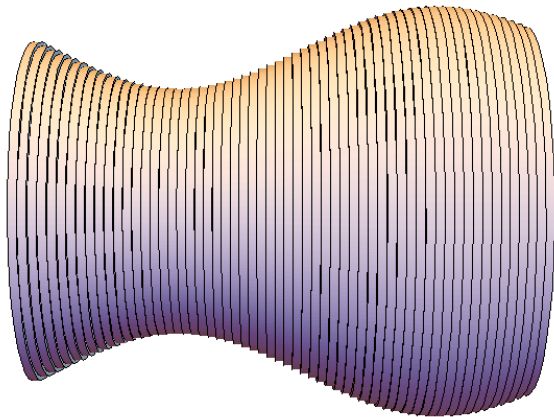
Volume por seção transversal



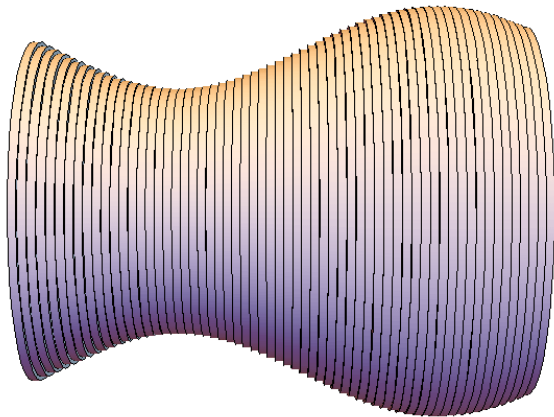
Volume por seção transversal



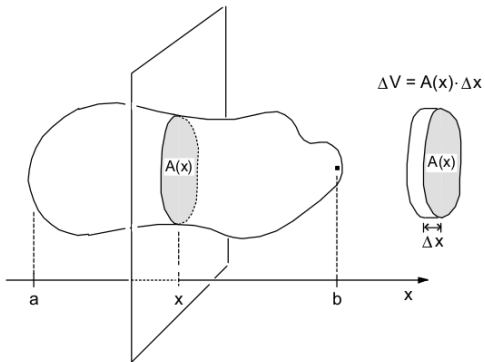
Volume por seção transversal



Volume por seção transversal

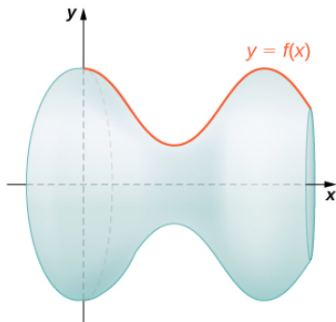
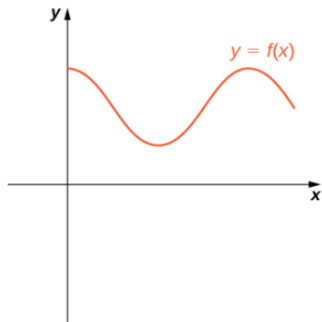


Volume por seção transversal

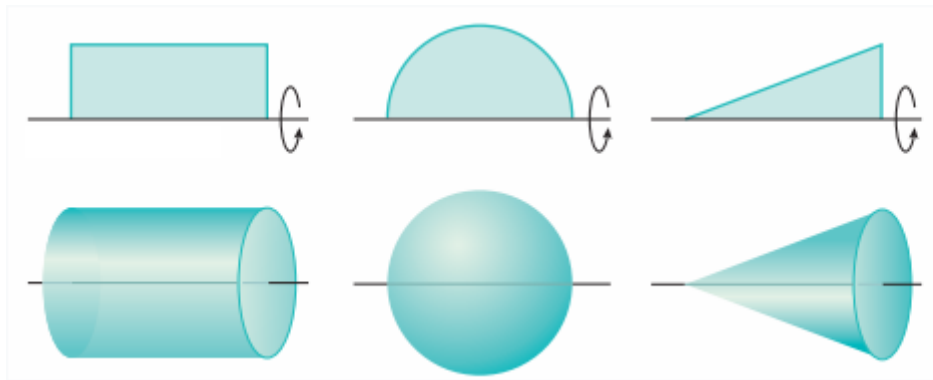


$$V = \int_a^b A(x) dx$$

Volume sólidos de revolução



Volume sólidos de revolução



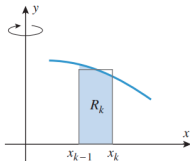
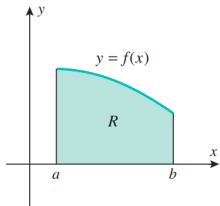
Volume sólidos de revolução

Caso particular do anterior.

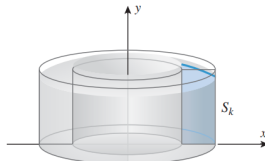
$$\text{Área da seção} = \pi f(x)^2.$$

$$\text{Volume} = \int_a^b \pi f(x)^2 dx.$$

Volume por cascas cilíndricas

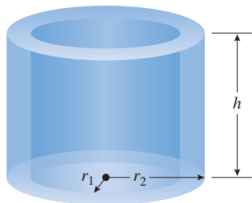


Retângulo aproximando
a k -ésima faixa.



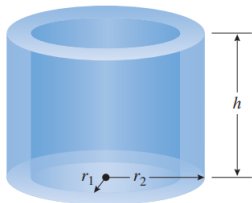
Camada cilíndrica
gerada pelo retângulo.

Volume por cascas cilíndricas

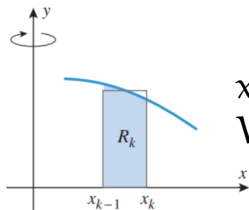


$$\begin{aligned} V &= [\text{área da seção transversal}] \cdot [\text{altura}] \\ &= (\pi r_2^2 - \pi r_1^2)h \\ &= \pi(r_2 + r_1)(r_2 - r_1)h \\ &= 2\pi \cdot \left[\frac{1}{2}(r_1 + r_2)\right] \cdot h \cdot (r_2 - r_1) \end{aligned}$$

Volume por cascas cilíndricas



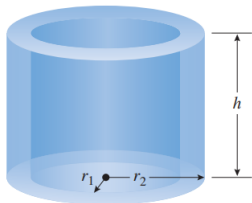
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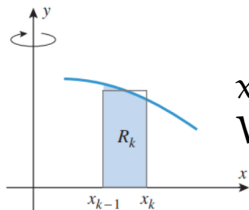
$x_k^* \in [x_{k-1}, x_k]$ ponto médio

$$V_k \approx 2\pi \cdot x_k^* \cdot f(x_k^*) \cdot \Delta x_k$$

Volume por cascas cilíndricas



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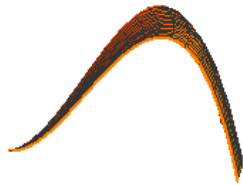
$$V_k \approx 2\pi \cdot x_k^* \cdot f(x_k^*) \cdot \Delta x_k$$

$$V = \int_a^b 2\pi x f(x) \, dx$$

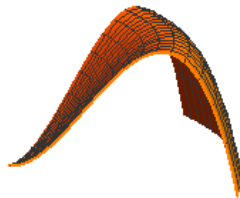
Volume por seção transversal



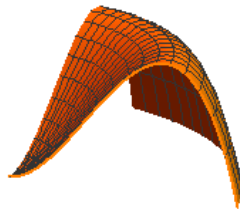
Volume por seção transversal



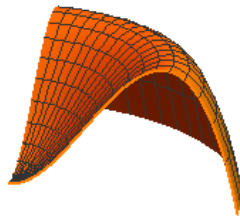
Volume por seção transversal



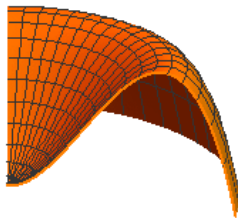
Volume por seção transversal



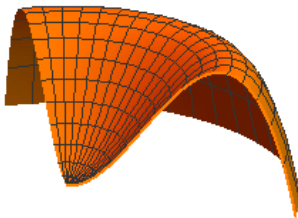
Volume por seção transversal



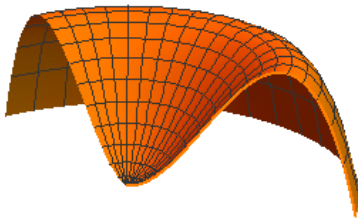
Volume por seção transversal



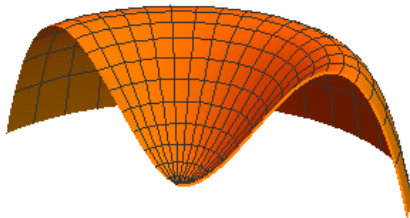
Volume por seção transversal



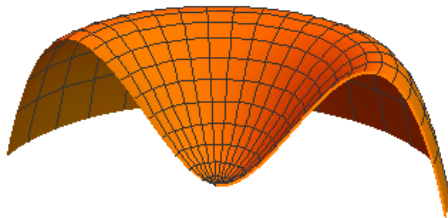
Volume por seção transversal



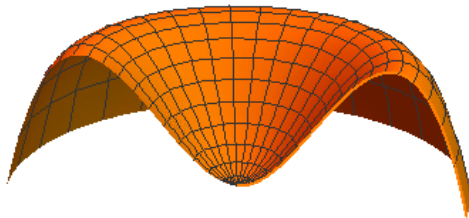
Volume por seção transversal



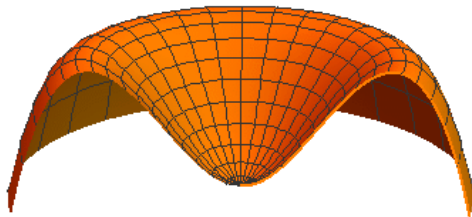
Volume por seção transversal



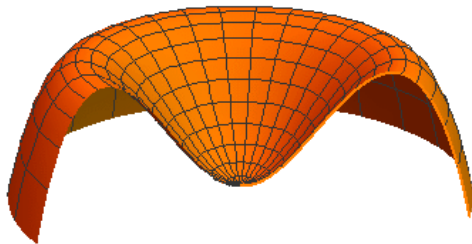
Volume por seção transversal



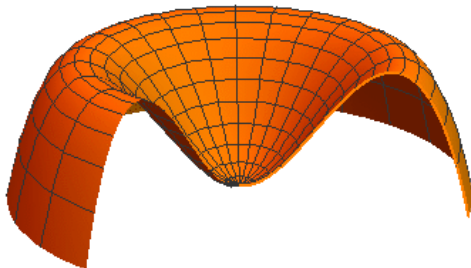
Volume por seção transversal



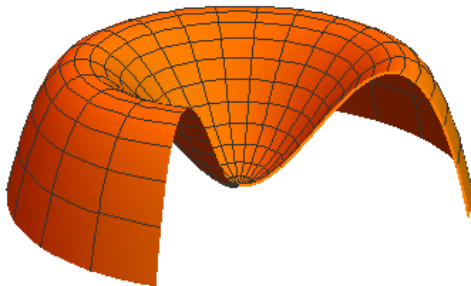
Volume por seção transversal



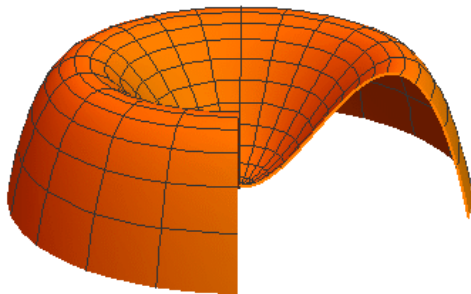
Volume por seção transversal



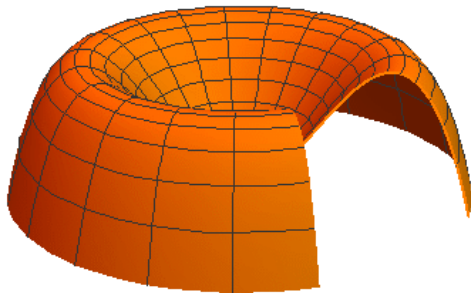
Volume por seção transversal



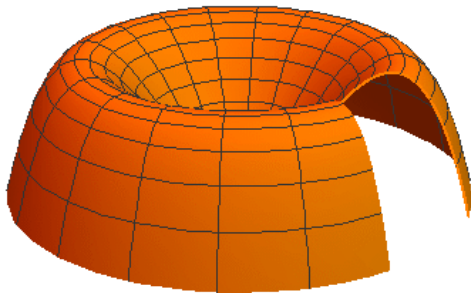
Volume por seção transversal



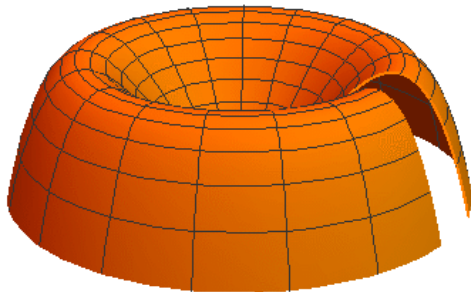
Volume por seção transversal



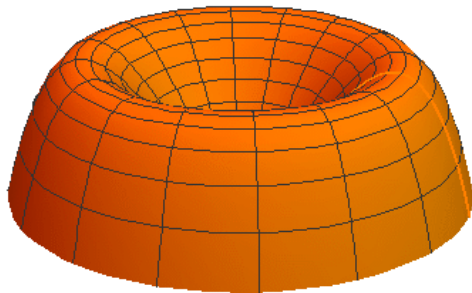
Volume por seção transversal



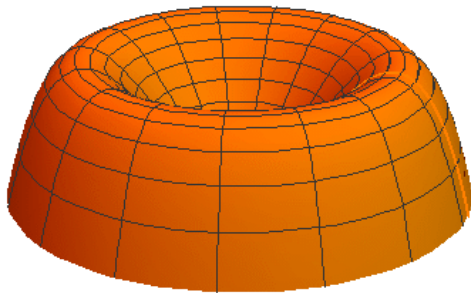
Volume por seção transversal



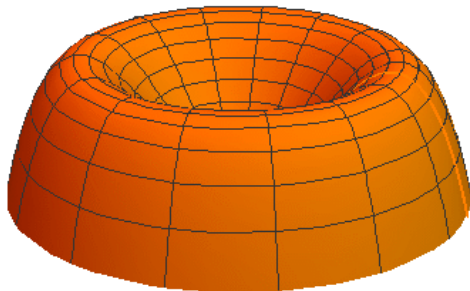
Volume por seção transversal



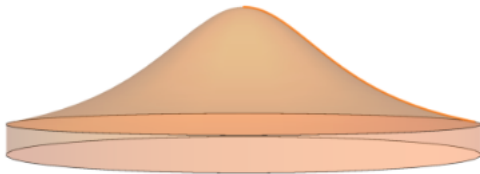
Volume por seção transversal



Volume por seção transversal



Determine o volume do sólido formado pela rotação da região limitada por $y = 0$, $x = 0$, $x = 2$ e $y = (1 + x^2)^{-1}$ ao redor do eixo y .



Integrais trigonométricas

$$\int \text{sen}^m(x) \cos^n(x) dx$$

1. $\int \cos^2(x) dx = \frac{1}{2} (x + \text{sen}(x) \cos(x)) + k.$

2. $\int \text{sen}^3(x) dx = \frac{1}{12} (-9 \cos(x) + \cos(3x)) + k.$

INTEGRAÇÃO DE PRODUTOS DE SENOS E COSSENOS

$\int \sin^m x \cos^n x \, dx$	PROCEDIMENTO	IDENTIDADES RELEVANTES
n ímpar	• Separe um fator de $\cos x$ • Aplique a identidade relevante • Faça a substituição $u = \sin x$	$\cos^2 x = 1 - \sin^2 x$
m ímpar	• Separe um fator de $\sin x$ • Aplique a identidade relevante • Faça a substituição $u = \cos x$	$\sin^2 x = 1 - \cos^2 x$
$\begin{cases} m \text{ par} \\ n \text{ par} \end{cases}$	• Use a identidade relevante para reduzir as potências de $\sin x$ e $\cos x$	$\begin{cases} \sin^2 x = \frac{1}{2}(1 - \cos 2x) \\ \cos^2 x = \frac{1}{2}(1 + \cos 2x) \end{cases}$